

HIRE-TO-RETIRE ANALYTICS

People Analytics Dataset — Data Dictionary & Business Guide

Version 1.0 | Synthetic Dataset | N = 750 Employees | 2019–2024

1. Executive Overview

This document provides the definitive reference for the **Hire-to-Retire Analytics Dataset**, a high-fidelity synthetic longitudinal dataset covering 750 simulated employees across the full employee lifecycle — from initial job requisition through post-exit analysis. The dataset is engineered for **People Analytics** use cases including workforce planning, talent quality measurement, retention modelling, compensation equity analysis, and manager effectiveness diagnostics.

The dataset is organized around four relational tables linked by **Candidate_ID** and **Employee_ID**. Each table maps to a distinct pillar of the employee lifecycle and can be queried independently or joined for end-to-end analysis.

Design Philosophy

Quality (Primary) › Cost (Secondary) › Speed (Tertiary). Every design decision reflects this hierarchy:

- Quality is embedded as a three-dimensional latent construct (Skill, Culture, Potential) that drives downstream performance, retention, and compensation outcomes.
- Cost is made explicit through Opportunity_Cost and Sourcing_Cost fields, allowing analysts to quantify the financial consequence of extended time-to-fill.
- Speed is deliberately subordinated: Days_to_Fill ranges 60–120 days, creating a visible, significant Opportunity Cost that challenges organisations to weigh hiring pace against hiring quality.

Key Metrics at a Glance

Metric	Value	Notes
Total Employees (N)	750	Synthetic; representative of a mid-size organisation

Metric	Value	Notes
Observation Period	2019–2024	6 years of longitudinal data
Avg Days to Fill	~90 days	Intentionally long — quality-prioritised hiring
Avg Opportunity Cost per Hire	~\$66,600 USD	At \$185K avg revenue/employee / 250 working days
Performance Correlation (r)	0.806	Quality_Index ↔ Avg KPI — within target 0.6–0.8 range
Performance Timeseries Records	~7,500	8–12 quarterly observations per employee
Termination Rate	~30–45%	Varies; driven by culture, KPI, compensation, manager quality
Biased Manager Rate	~5%	Manager pool of 60; ~3 flagged as carrying interviewer bias
Poor Manager Rate	~10%	~6 managers known to suppress high-quality employee outcomes

2. Relational Schema & Table Overview

The four tables form a hub-and-spoke relational model. Recruitment is the anchor (Pillar 1); all other tables extend from it using shared key fields.

Table	Primary Key	Foreign Keys	Rows	Cols	Lifecycle Pillar
Recruitment	Candidate_ID	—	750	13	Pillar 1 — Talent Acquisition Anchor
Employee_Master	Employee_ID	Candidate_ID	750	7	Pillar 2 — Core HR Record
Performance_Timeseries	Employee_ID + Quarter	Employee_ID, Candidate_ID	~7,500	8	Pillars 2 & 4 — Performance Management
Lifecycle_Retention	Employee_ID	Candidate_ID	750	11	Pillar 3 — Engagement & Retention

3. The Quality Matrix — Core Analytical Engine

The **Quality Matrix** is the dataset's central analytical construct. Every employee carries three latent quality scores assessed at point-of-hire. These are not observed directly by managers — they represent the **true underlying capability profile** that analytics practitioners are trying to infer from downstream signals.

Dimension	Field	What It Measures	Distribution	Key Downstream Link
Skill Score	Skill_Score	Technical proficiency, domain expertise, and hard skills. The capacity to perform the core job functions from day one.	Beta(5,3)×100 — right-skewed, most hires competent	KPI_Achievement, Quality_of_Work
Culture Score	Culture_Score	Alignment with organisational values, communication style, teamwork, and soft skills. Determines 'fit' beyond credentials.	Beta(4,3)×100 — moderate right skew	Avg_Monthly_Pulse, Termination_Date
Potential Score	Potential_Score	Future-readiness: capacity for growth, leadership, and skill development. Predicts long-term value creation.	Beta(3,3)×100 — near-symmetric	Skill_Acquisition_Score, Promotion_Count
Quality Index	Quality_Index	Composite mean of the three dimensions. Primary predictor of overall performance outcomes.	Mean of three Beta draws	KPI_Achievement (r ≈ 0.81)

Important note for analysts: The Quality Matrix scores are ground-truth latent variables. Regression models predicting KPI from these scores will show strong, clean signal. To simulate a **real-world discovery exercise**, analysts should attempt to **reconstruct** quality scores from observed performance data — the noise and management effects in the dataset will make this a realistic challenge.

4. Opportunity Cost Logic

Every unfilled requisition represents **lost revenue-generating capacity**. The **Opportunity_Cost** field quantifies this daily leakage.

$$\text{Opportunity_Cost} = \text{Days_to_Fill} \times (\text{Avg_Revenue_Per_Employee} \div 250)$$

Default parameters: Average Revenue Per Employee = \$185,000 | Working days = 250 | Days to Fill = 60–120.

At these parameters, a single hire with a 90-day time-to-fill incurs approximately **\$66,600 in foregone revenue**. Across all 750 hires, total opportunity cost in the dataset typically exceeds **\$49 million** — a figure that makes the business case for quality-optimised hiring vividly concrete.

Analytical use: Segment Opportunity_Cost by Sourcing_Channel to determine whether premium channels (Agency, LinkedIn) justify their higher Sourcing_Cost through faster fill times. Calculate total cost-of-acquisition as Sourcing_Cost + Opportunity_Cost for true channel ROI.

5. Full Data Dictionary

All field definitions, data types, business rules, and formulas are documented below by table.

5.1 Table: Recruitment (Pillar 1 — Anchor)

Field Name	Data Type	Description	Business Rule / Formula
Candidate_ID	String	Unique identifier for each candidate/applicant. Serves as the primary key linking all four tables.	Format: CAND-XXXXXX. Immutable across lifecycle.
Employee_ID	String	Assigned upon hire. Links the recruitment record to all downstream employee tables.	Format: EMP-XXXXXX. Null until offer accepted.
Req_ID	String	Unique job requisition identifier associated with the open position.	Format: REQ-XXXXXX. One req per hire in this dataset.
Sourcing_Channel	String	The channel through which the candidate was sourced (e.g., LinkedIn, Agency, Referral).	Categorical. Affects Sourcing_Cost and, under bias, downstream scores.
Sourcing_Cost	Float	Monetary cost (USD) of acquiring the candidate through the given channel.	Base cost per channel $\pm$ 30% random variance.
Vacancy_Start_Date	Date	The date the job requisition was officially opened.	ISO date format (YYYY-MM-DD).
Hire_Date	Date	The date the candidate formally accepted and started employment.	Vacancy_Start_Date + Days_to_Fill.
Days_to_Fill	Integer	Number of calendar days from requisition open to hire. Intentionally long (60–120 days) reflecting quality-prioritized hiring.	Range: 60–120 days. Core driver of Opportunity_Cost.
Skill_Score	Float	Latent quality dimension: technical proficiency and domain expertise. Scored 0–100.	Sampled from Beta(5,3) $\times$ 100. Primary predictor of KPI_Achievement and Quality_of_Work.
Culture_Score	Float	Latent quality dimension: alignment with organizational values and soft skills. Scored 0–100.	Sampled from Beta(4,3) $\times$ 100. Drives

Field Name	Data Type	Description	Business Rule / Formula
			Pulse_Score decay rate and retention.
Potential_Score	Float	Latent quality dimension: future growth capacity and leadership readiness. Scored 0–100.	Sampled from Beta(3,3)×100. Primary predictor of Skill_Acquisition_Score and promotions.
Quality_Index	Float	Composite latent quality metric — simple mean of the three quality dimensions.	(Skill_Score + Culture_Score + Potential_Score) / 3. Correlates $r \approx 0.8$ with avg KPI.
Opportunity_Cost	Float	Estimated revenue foregone (USD) while the position remained vacant.	Days_to_Fill × (Avg_Revenue_Per_Employee / 250).

5.2 Table: Employee_Master (Pillar 2)

Field Name	Data Type	Description	Business Rule / Formula
Employee_ID	String	Primary key for all post-hire tables. Assigned at hire.	Format: EMP-XXXXX.
Candidate_ID	String	Foreign key linking to the Recruitment table.	Joins to Candidate_ID in Table 1.
Hire_Date	Date	Repeated from Recruitment for convenience. The employee start date.	Matches Hire_Date in Table 1.
Role	String	Job title within the employee's department.	Sampled from department-specific role list.
Department	String	Business unit the employee belongs to (e.g., Engineering, Sales, HR).	10 departments: Engineering, Sales, Finance, HR, Operations, Marketing, Legal, Customer Success, Product, Data.
Base_Salary	Integer	Annual base compensation (USD), rounded to nearest \$1,000.	Driven by department salary band and Skill_Score. Range: \$60K–\$210K.

Field Name	Data Type	Description	Business Rule / Formula
Manager_ID	String	Identifier of the employee's direct manager (pool of 60 managers).	~5% flagged as biased managers; ~10% as poor managers. See bias & noise documentation.

5.3 Table: Performance_Timeseries (Pillars 2 & 4)

Field Name	Data Type	Description	Business Rule / Formula
Employee_ID	String	Foreign key. Joins to Employee_Master.	One employee has 8–12 quarterly records.
Quarter	String	Calendar quarter label (e.g., Q2-2021).	Derived from Hire_Date + tenure quarter index.
Tenure_Quarter	Integer	Ordinal quarter number since hire (1 = first full quarter).	Range: 1–12.
KPI_Achievement	Float	Percentage of quarterly KPI targets met. Core performance metric. Scored 0–100.	$f(\text{Quality_Index}, \text{Tenure}, \text{Mgmt_Noise}, \text{Culture_Save})$. $r \approx 0.70$ with Quality_Index.
Quality_of_Work	Float	Subjective quality rating of output. Scored 0–100.	$f(\text{Skill_Score}, \text{Tenure})$. $r \approx 0.65$ with Skill_Score.
Skill_Acquisition_Score	Float	Rate of new skill development in the quarter. Scored 0–100.	$f(\text{Potential_Score}, \text{Tenure})$. $r \approx 0.60$ with Potential_Score.
Mgmt_Noise_Applied	Boolean	Flag: TRUE if a poor manager artificially deflated this employee's performance scores.	Affects ~8% of high-quality hires assigned to poor managers.
Culture_Save_Applied	Boolean	Flag: TRUE if a high Culture_Score compensated for low overall quality to sustain performance.	Affects ~6% of low-quality hires with Culture_Score > 70.

5.4 Table: Lifecycle_Retention (Pillar 3)

Field Name	Data Type	Description	Business Rule / Formula
Employee_ID	String	Foreign key. Joins to Employee_Master.	Primary key for lifecycle tracking.
Candidate_ID	String	Foreign key linking back to Recruitment table.	Enables full lifecycle tracing from sourcing to exit.
Avg_Monthly_Pulse	Float	Mean of simulated monthly engagement/pulse survey scores over tracked period. Scored 0–100.	Base = 50 + (Culture_Score × 0.4). Decays faster when Culture_Score < 70.
Last_Pulse_Score	Float	Final observed monthly pulse score (most recent month). Early warning retention indicator.	Values < 40 significantly increase Termination probability.
Promotion_Count	Integer	Total number of promotions received during employment.	Binomial draw. p driven by Potential_Score and avg KPI. Max 3.
Promotion_Dates	String	Semicolon-delimited list of promotion event dates, or 'None'.	Dates derived from Hire_Date + random interval (1–4 years per event).
Total_Comp_vs_Market	Float	Ratio of employee's total compensation to external market median. >1.0 = above market.	Baseline N(1.0, 0.1). Biased managers reduce this by 5–15% for certain sourcing channels.
Months_Tracked	Integer	Number of months for which pulse data was collected.	Range: 12–36 months.
Terminated	Boolean	Indicates whether the employee has left the organization.	Probability function of Culture_Score, KPI, Pulse, Comp_vs_Market, and Manager type.
Termination_Date	String	Date of exit if terminated, or 'Active' if still employed.	'Active' for current employees. Date otherwise.
Exit_Reason	String	Reason for departure (voluntary, involuntary, retirement) or 'Active'.	Involuntary-Performance if avg KPI < 40; otherwise probability-weighted voluntary/involuntary.

6. Realistic Noise & Bias Mechanisms

The dataset deliberately embeds noise and bias to mirror real-world complexity. Treating these as modelling targets — rather than cleaning them away — is the intended use.

6.1 Management Effect (Poor Manager Noise)

Approximately **10% of managers (6 of 60)** are flagged as poor managers. High-quality hires (Quality_Index > 70) assigned to these managers experience a **15–30 point suppression** on their KPI_Achievement scores. This models the real-world phenomenon where organisational dysfunction — not talent — is the root cause of underperformance.

- The Mgmt_Noise_Applied flag (TRUE/FALSE) in Performance_Timeseries identifies affected records.
- Analytical challenge: Build a model to identify poor managers by detecting systematic KPI suppression in their direct reports, controlling for employee Quality_Index.

6.2 Culture Survival Effect

Approximately **6% of low-quality hires** (Quality_Index < 40) with high Culture_Scores (> 70) receive a **8–18 point performance boost**. This models employees who thrive through relationship capital, institutional knowledge, and team cohesion even when technical skills are limited.

- The Culture_Save_Applied flag in Performance_Timeseries identifies these records.
- Key insight: Culture_Score is a stronger predictor of retention than either Skill_Score or Potential_Score — a common real-world finding.

6.3 Interviewer Bias (5% Manager Effect)

Approximately **5% of managers (~3 of 60)** carry a bias flag. These managers consistently **underscore candidates sourced from 'non-traditional' channels** (Job Board, Direct Apply, University) across two outcome dimensions:

- Performance: KPI_Achievement and Quality_of_Work are suppressed by 5–12 and 3–9 points respectively in Performance_Timeseries.
- Compensation: Total_Comp_vs_Market is reduced by 5–15% in Lifecycle_Retention for affected employees.

Analytical use: A bias audit should compare KPI and comp outcomes by Sourcing_Channel, segmented by Manager_ID. Biased managers will show a statistically significant gap between channel cohorts that does not appear in unbiased managers' teams.

7. Retention & Termination Logic

Termination probability is a composite function of five factors. Understanding the relative weights helps interpret retention models built on this data.

Factor	Direction	Magnitude	Business Interpretation
Culture_Score	Low → ↑ Termination	Primary	Low-culture employees disengage faster. The pulse score decay mechanism makes this visible over time.
Avg KPI_Achievement	Low → ↑ Termination	Primary	Consistently poor performance triggers involuntary exit pathways.
Last_Pulse_Score < 40	True → ↑ Termination	+15% prob	Late-stage disengagement is the strongest near-term flight risk signal.
Total_Comp_vs_Market < 0.90	True → ↑ Termination	+10% prob	Below-market compensation is a leading indicator of voluntary attrition.
Poor Manager Assignment	True → ↑ Termination	+10% prob	Manager quality independently drives attrition, separate from employee quality.

Pulse Score Decay Model: Monthly Pulse Score is calculated as **Base_Pulse – (decay_rate × month_index × 10) + noise**. The decay rate = max(0.1, (70 – Culture_Score) / 300). Employees with Culture_Score = 30 decay at ~0.13/month; employees with Culture_Score = 80 decay at ~0.03/month. This creates the observable **culture-retention correlation** that a well-built survival analysis model should recover.

8. Recommended Analytical Use Cases

The following analyses represent the primary intended applications of this dataset, roughly ordered from descriptive to predictive complexity.

8.1 Sourcing Channel ROI Analysis

Join: Recruitment only. Calculate total cost-of-acquisition = Sourcing_Cost + Opportunity_Cost. Rank channels by **Quality_Index per dollar spent**. Employee Referrals typically dominate on quality-adjusted ROI; Agencies have high absolute cost but may fill faster (lower Opportunity_Cost).

8.2 Quality-Performance Predictive Model

Join: Recruitment ↔ Performance_Timeseries (via Employee_ID). Build a regression predicting KPI_Achievement from the three Quality Matrix scores plus Tenure_Quarter. Expected result: $r^2 > 0.50$, with Mgmt_Noise as the key unexplained residual.

8.3 Retention Survival Analysis

Join: All four tables. Construct a Cox proportional hazards model with Termination_Date as the event time. Covariates: Culture_Score, Avg_Monthly_Pulse, Total_Comp_vs_Market, Manager_ID (fixed effect), and Department. Culture_Score and Last_Pulse_Score should emerge as dominant hazard factors.

8.4 Manager Effectiveness Scorecard

Join: Employee_Master ↔ Performance_Timeseries ↔ Lifecycle_Retention. For each Manager_ID, compute mean team KPI controlling for avg Quality_Index, team attrition rate, and avg pulse score. Outlier managers with low team performance despite high-quality subordinates signal the poor-manager population.

8.5 Compensation Equity Audit

Join: Recruitment ↔ Employee_Master ↔ Lifecycle_Retention. Regress Total_Comp_vs_Market on Sourcing_Channel, controlling for Department, Base_Salary, and Tenure. Biased Manager_IDs will produce a statistically significant coefficient on 'Job Board' and 'Direct Apply' channels, modelling a compensation disparity flag.

8.6 Workforce Planning Cost Simulation

Join: Recruitment only. Simulate the impact of reducing Days_to_Fill by 15 days across all future hires. Multiply by (Avg_Revenue_Per_Employee / 250) × 15 × projected headcount to quantify the Opportunity_Cost savings. Balance against the quality degradation that faster hiring might introduce.

9. Parameterisation & Customisation Guide

The Python generation script is modular. The following parameters can be adjusted at the top of the script to alter dataset characteristics without modifying core logic.

Parameter	Default	Effect of Increasing
N_EMPLOYEES	750	Larger dataset; more statistical power but longer runtime.
AVG_REVENUE_PER_EMP	\$185,000	Increases Opportunity_Cost values across all requisitions.
MIN/MAX_DAYS_TO_FILL	60 / 120	Widens or narrows the cost range; reduces to 30–60 for a speed-prioritised scenario.
QUALITY_PERF_CORR	0.70	Strengthens the quality-performance link; makes predictive models cleaner.
BIAS_MANAGER_FRAC	5%	More managers carry bias; bias signal becomes easier to detect.

Parameter	Default	Effect of Increasing
NOISE_MGMT_FAIL_FRAC	8%	More high-quality hires fail due to management; complicates quality-outcome correlation.
NOISE_CULTURE_SURVIVE_FRAC	6%	More low-quality hires survive via culture; muddies predictive models constructively.

10. Glossary of People Analytics Terms

Term	Definition
Latent Quality Profile	A multi-dimensional construct representing an employee's true underlying capability, not directly observable but inferred from downstream performance signals.
Opportunity Cost of Vacancy	The revenue foregone while a productive role remains unfilled. Calculated as daily revenue rate × days unfilled.
Pulse Score	A periodic (monthly) micro-survey metric capturing employee engagement and job satisfaction. Leading indicator of attrition risk.
Comp vs Market Ratio	Ratio of an employee's total compensation to the median external market rate for the same role/level. Values below 0.90 are strong attrition predictors.
Time to Fill	Calendar days between requisition opening and offer acceptance. Primary driver of Opportunity Cost.
Quality-Adjusted ROI	A recruiting channel effectiveness metric that normalises Quality_Index achieved per dollar of sourcing cost.
Survival Analysis	A class of statistical models (e.g., Kaplan-Meier, Cox regression) for modelling time-to-event data such as employee tenure and exit timing.
Hazard Rate	In survival analysis, the instantaneous probability of the termination event occurring at time t, given survival to that point.
Interviewer Bias	Systematic scoring distortion by assessors, favouring or disfavouring candidates based on characteristics unrelated to job performance (e.g., sourcing channel as a proxy for socioeconomic background).
Hire-to-Retire Lifecycle	The complete arc of an employee's relationship with an organisation: from job requisition, through recruitment, onboarding, performance, development, to eventual exit.
Management Noise	Performance variance attributable to manager quality rather than employee capability. A key confounder in people analytics models.

11. Data Limitations & Caveats

- This is entirely synthetic data. All names, IDs, and figures are simulated. Do not represent as real employee records.
- The Quality Matrix scores are not observable in practice. Real-world practitioners must infer quality from assessments, reference checks, and early performance signals.
- The bias mechanism is a simplification — it operates on sourcing channel as a proxy rather than individual demographics, to avoid generating sensitive demographic data.
- Performance scores are quarterly aggregates. Real performance systems contain richer nuance including 360 feedback, project-level output, and manager narrative.
- The dataset does not model leave events (parental, medical, sabbatical) which meaningfully affect performance timeseries in practice.
- Compensation data (Base_Salary) does not include bonus, equity, or benefits — key components of total rewards that affect retention differently.